

SURVEY

Microcontroller-Driven Battery Management in Hybrid Energy Systems: A Systematic Review of Applications, Control Strategies, and Emerging Trends

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This work was supported by the University of South Africa, Roodepoort, South Africa.

ABSTRACT Microcontroller-driven battery management systems (BMS) are crucial for various applications, including electric vehicles, portable electronics, and renewable energy storage. These systems monitor and control critical parameters such as voltage, current, temperature, and state of charge to optimize battery performance and lifespan. Hence, this study aims to conduct a systematic literature review on microcontroller-driven battery management in hybrid energy systems by focusing on applications, control strategies, and emerging trends, among others. To collect documents, a systematic procedure was employed using targeted keywords to search within the Scopus database, while focusing on relevant, recent engineering papers. Eligible documents were refined, exported, and carefully assessed for relevancy, ensuring a comprehensive dataset for analysis. Among others, review findings highlighted that microcontroller-based systems are making big strides in renewable energy and battery management. Tools like Maximum Power Point Tracking (MPPT) and Pulse Width Modulation (PWM) help improve efficiency, while microcontrollers like Arduino and TMS320F28379D balance cost and performance depending on the project. Lithium-ion batteries are favored for their long life, but lead-acid batteries still have their place because they are cheaper. This study aims to contribute by conducting a systematic literature review using tailored keywords to identify and analyze advancements in microcontroller-driven battery management systems. Key contributions include reviewing innovations in battery charging systems, exploring hybrid systems integrating multiple power sources, examining control strategies and algorithms, and assessing renewable energy-based charging systems. The study further identifies patterns, themes, and emerging trends in the literature, highlights research gaps, and proposes directions for future work in this field.

INDEX TERMS Battery management systems, control strategies and algorithms, hybrid energy systems, microcontroller, renewable energy.

I. INTRODUCTION

Microcontrollers are integrated circuits designed for specific applications, containing essential components of a computer system including CPU, memory, and I/O capabilities [1],

The associate editor coordinating the review of this manuscript and approving it for publication was Amin Mahmoudi¹.

[2]. They are crucial in automatically controlled devices, from automotive systems to household appliances, due to their ability to process input and execute specific actions [3], while they also operate at lower speeds and consume less power compared to general-purpose computers, making them ideal for embedded. Programming microcontrollers involves using various software tools and can be challenging for

those without specialized skills [4]. Recent developments allow programming in high-level languages, enabling complex control algorithms [5]. The market offers a variety of microcontrollers with different structures and capabilities, including single-board computers like Arduino and Raspberry Pi [6]. Understanding microcontroller concepts is essential for selecting appropriate hardware for specific applications [7].

Microcontroller-driven battery management systems (BMS) are crucial for various applications, including electric vehicles, portable electronics, and renewable energy storage [8], [9]. These systems monitor and control critical parameters such as voltage, current, temperature, and state of charge to optimize battery performance and lifespan [10], [11]. Low-cost microcontrollers, like Arduino, can efficiently measure discharge current and time to estimate battery capacity [12]. BMS designs often incorporate modules for measurement, alarm, input/output, and communication [13]. Some advanced features include cell balancing, which can be implemented through passive or active methods [11], [14]. Data logging and wireless communication capabilities allow for real-time monitoring and performance analysis [10], [14]. General, microcontroller-driven BMS solutions offer efficient, cost-effective ways to maximize battery storage capacity and cycle life across various applications.

They also play a crucial role in hybrid energy systems, optimizing power flow and extending battery life. These systems can effectively control charging and discharging processes, select appropriate energy sources, and protect against excessive currents [15]. Implementations have shown that hybrid energy sources can significantly reduce charging times compared to single sources [16]. Microcontrollers enable accurate monitoring of battery parameters such as temperature, electrolyte level, and backup hours [8]. They also facilitate versatile power transfer and improved power quality management in smart microgrids [17]. Various control strategies, including hysteresis-based approaches, have been developed to optimize system operation [18], [19]. These systems have applications in electric vehicles, renewable energy storage, and residential settings, where they can monitor state of charge, temperature, and current [9].

A range of literature reviews have explored the use of microcontrollers for battery discharging and charging in hybrid energy systems. Parthasarathy and Vijayaraj [20] and Zainurin et al. [21] both highlight the potential of microcontrollers, with Parthasarathy and Vijayaraj specifically focusing on the use of an Arduino Nano for battery charge control in a solar-powered system. Banguero et al. [22] and Hariprasad et al. [23] discuss the importance of control strategies and Battery Management Systems (BMS) in renewable energy systems, with Hariprasad emphasizing the role of microcontrollers in monitoring and controlling battery parameters. Kalaivani et al. [24] and Arrar and Xioaning [25] both delve into the use of microcontrollers in hybrid microgrid systems, with Kalaivani focusing on power management

while Arrar and Xioaning on energy management using smart controllers. Gajewski, and Pienkowski [26] and Coppez et al. [27] provide broader overviews of hybrid renewable energy systems and energy storage needs, with Gajewski discussing the control of a hybrid renewable energy system and Coppez addressing the challenges and opportunities in integrating batteries with renewable energy systems.

Various studies have highlighted the potential of microcontrollers in related contexts such as battery charge control in solar-powered systems, control strategies and BMS in renewable energy systems, and energy and power management in hybrid microgrid systems. However, none have provided a focused review on microcontroller applications for both charging and discharging batteries within hybrid energy systems. Thus, this study aims to conduct a systematic literature review on microcontroller-driven battery management in hybrid energy systems by focusing on applications, control strategies, and emerging trends. The following objectives and contributions are intended to be achieved in this study.

- Develop keywords “battery discharge” OR “battery charging” AND “hybrid system” OR “Microcontroller,” and conduct a systematic literature review.
- Review microcontroller-based innovations in battery charging systems.
- Explore hybrid systems that integrate multiple power sources.
- Examine currently available control strategies and algorithms for battery management systems.
- Review renewable energy-based charging systems
- Identification of patterns, themes, and emerging trends from existing literature.
- Research gap identification and future research direction proposals

The remaining parts of the study include section II, which presents the conceptual framework for microcontroller applications in battery management for hybrid systems, the section offers an overview of hybrid energy systems and battery management, section IV detailing the methodology used in this study, section V providing a literature review synthesis, section VI focusing on results and discussions, including pattern identification and analysis, section VII addressing gap identification and future research directions, and finally, section VIII presenting the study conclusions.

II. CONCEPTUAL FRAMEWORK FOR MICROCONTROLLER APPLICATIONS IN BATTERY MANAGEMENT FOR HYBRID SYSTEMS

A. CONCEPTUAL FRAMEWORKS IN BATTERY MANAGEMENT SYSTEMS

BMS is crucial for optimizing the performance, safety, and longevity of lithium-ion batteries in various applications. Conceptual frameworks for BMS incorporate cloud computing, machine learning, and artificial intelligence to enhance real-time monitoring, analysis, and decision-making [28].

A multilayer design architecture has been proposed, focusing on physical basis, algorithmic understanding, and application management [29]. BMS frameworks can utilize goal-seeking approaches for charge and discharge operations [30] and employ advanced techniques like Unscented Kalman Filters for accurate state estimation [31]. Safety-critical applications require hardware and software isolation, redundancy, and partitioning [32]. BMS technologies encompass battery monitoring, diagnostics, charge control, state estimation, protection, and thermal management [33]. Future BMS development trends focus on data-driven intelligence, improved safety, and interconnectivity [29].

B. INTEGRATING THEORY AND PRACTICE IN MICROCONTROLLER APPLICATIONS

The integration of theory and practice in microcontroller applications is a crucial challenge across various disciplines. Several papers highlight the need for bridging this gap through conceptual frameworks and innovative teaching methodologies [34], [35], [36]. Modern programming practices and design patterns are essential for advancing microcontroller application development, particularly in embedded systems [37]. Bridging theory and practice is a key aspect of human cognition, with expertise enabling better situation recognition and decision-making [38].

Researchers have developed frameworks to bridge the gap between theoretical knowledge and practical application, highlighting the importance of training methods that integrate the two [39]. In control systems, the gap between theory and practice poses a significant challenge, highlighting the need for targeted research and practical solutions to address it [40]. Resources like textbooks remain essential for helping students and professionals understand and apply microcontroller technologies effectively [41].

C. FUTURE DIRECTIONS USING THE CONCEPTUAL FRAMEWORK

The conceptual framework for hybrid energy systems (HES) guides future research and development in several key areas. It emphasizes the need for optimizing cost and performance through multi-stage decision-making processes [42], [43]. The framework incorporates various renewable energy sources, storage technologies, and integration strategies [44], while future directions include improving control and reliability approaches, exploring artificial intelligence applications, and focusing on localized integration.

The framework also addresses economic considerations, proposing broader theoretical models based on production and utility theory [45]. It supports the introduction of renewable energy and carbon emission constraints into existing power networks [46]. Additionally, the framework emphasizes modularity and flexibility to adapt to evolving energy systems [47] and considers the complex societal co-evolutionary processes involved in system transitions [48].

III. OVERVIEW OF HYBRID ENERGY SYSTEMS AND BATTERY MANAGEMENT

A. HYBRID ENERGY SYSTEMS

Hybrid energy systems, integrating renewable sources like solar and wind with traditional power systems, offer enhanced energy security, reduced greenhouse gas emissions, and cost efficiencies [49], [50]. However, their effective management, especially regarding battery storage, poses significant challenges [51]. Batteries, essential for storing surplus energy, must operate within optimal conditions to prevent overcharging and deep discharging, which can degrade performance and lifespan [52]. Advanced control algorithms and microcontroller-based management systems play a crucial role in monitoring and regulating battery operations, ensuring efficiency, longevity, and system stability [53], [54]. These systems can make real-time adjustments based on energy demand and supply fluctuations, enhancing overall energy utilization and system resilience.

B. USE OF MICROCONTROLLER

A Microcontroller Unit (MCU) is a compact integrated circuit designed to manage specific functions within an embedded system [55]. An MCU integrates one or more CPUs (processor cores), memory, and programmable input/output peripherals [56]. Unlike a microprocessor, which is a standalone processor that requires external memory, a microcontroller combines a processor with built-in (internal) memory, enabling it to function independently in embedded applications [57].

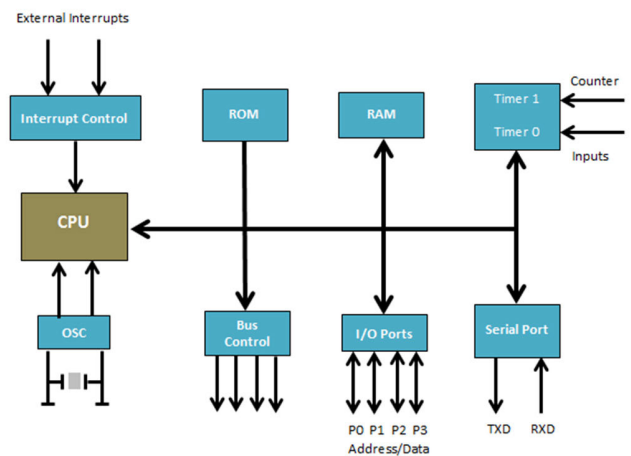


FIGURE 1. The internal architecture of a microcontroller unit.

Figure 1 illustrates the internal architecture of the microcontroller. The CPU orchestrates and synchronizes processes, manages the registers, and interprets ROM data. If a higher-priority program requests access to the system bus, an interrupt subroutine is invoked. Interrupts enable current processes to be delayed for this additional access to happen. The oscillator (denoted as OSC in the diagram) handles the timing operations for the microcontroller's digital circuits.

C. BATTERY DISCHARGING AND CHARGING

A battery is a device that stores chemical energy and converts it to electrical energy, playing pivotal roles in hybrid energy systems, electric vehicles, and renewable energy integration [58]. Battery discharging involves the controlled release of stored chemical energy from battery electrodes, converting it into electrical energy. Conversely, battery charging reverses this process, by restoring chemical potential through methods like constant current or voltage charging [59]. There are four types of batteries in the market that are commonly used as rechargeable batteries.

1) LEAD-ACID BATTERIES

The Lead-acid batteries are composed of electrodes made of lead dioxide for the positive plate and spongy lead for the negative plate, both immersed in a sulfuric acid electrolyte [60]. Each cell in a lead-acid battery typically provides a nominal voltage of 2.0 volts, with the current capacity varying widely, from a few milliamps in small-scale applications to several hundred amps in automotive and industrial applications, these batteries often consist of multiple cells connected in series to achieve the required voltage [61]. Figure 2 presents internal architecture of lead-acid batteries.

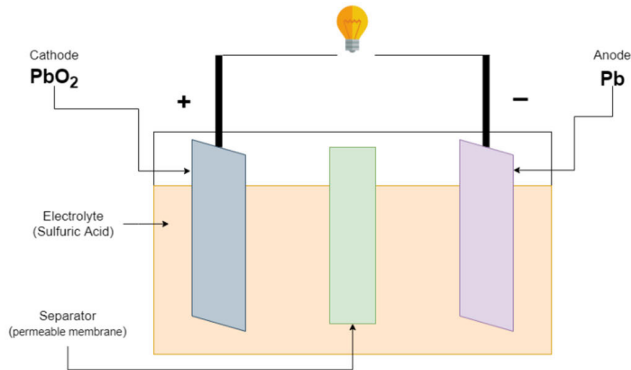
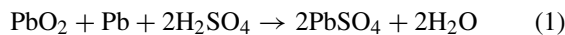
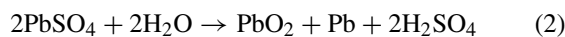


FIGURE 2. The internal architecture of lead-acid batteries.

Lead-acid batteries function through reversible chemical reactions between lead dioxide (PbO_2) and lead (Pb) in the presence of sulfuric acid (H_2SO_4). During discharging, lead dioxide at the positive electrode and lead at the negative electrode react with sulfuric acid, forming lead sulphate ($PbSO_4$) and water (H_2O), while releasing electrical energy. The reaction can be represented as:



When charging, this reaction is reversed. The lead sulphate and water convert back into lead dioxide, lead, and sulfuric acid, restoring the battery's chemical potential. The charging reaction is:



2) NI-CD BATTERIES

Nickel-cadmium batteries feature a nickel oxyhydroxide positive electrode, a cadmium negative electrode, and a

potassium hydroxide electrolyte, providing each cell with a nominal voltage of 1.2 volts and current capacities from a few hundred milliamps to several amps, depending on the application [62]. Ni-Cd batteries are known for their robustness and ability to deliver high-discharge currents. Figure 3 is the illustration of the internal architecture of Ni-Cd batteries.

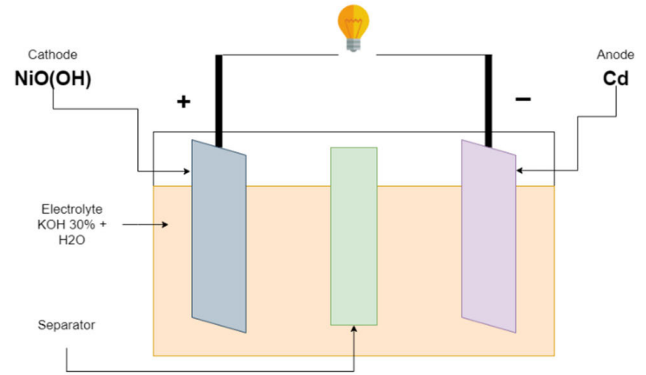
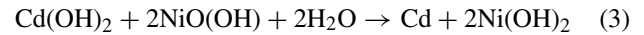
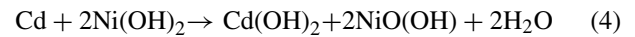


FIGURE 3. The internal architecture of Ni-Cd batteries.

Nickel-cadmium ($Ni - Cd$) batteries operate via reactions involving cadmium hydroxide ($Cd(OH)_2$) and nickel oxyhydroxide ($NiO(OH)$). During discharging, cadmium hydroxide at the positive electrode and nickel oxyhydroxide at the negative electrode react to form cadmium Cd and nickel hydroxide ($Ni(OH)_2$), releasing electrical energy. This process is described by:



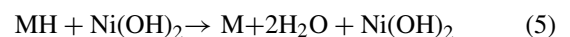
In the charging phase, cadmium and nickel hydroxide are converted back to cadmium hydroxide and nickel oxyhydroxide:



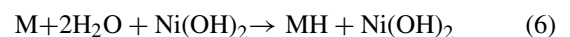
3) NI-MH BATTERIES

Nickel-metal hydride batteries feature a nickel oxyhydroxide positive electrode and a metal hydride negative electrode, with an alkaline electrolyte, usually potassium hydroxide. The nominal voltage of each cell is 1.2 volts, like Ni-Cd batteries, and they offer higher energy density with current ratings ranging from a few hundred milliamps to several amps [63]. Ni-MH batteries are commonly used in consumer electronics due to their improved capacity and reduced memory effect compared to Ni-Cd batteries [64]. Nickel-metal hydride ($Ni - MH$) batteries use a metal hydride (MH) as the positive electrode material and nickel hydroxide ($Ni(OH)_2$) at the negative electrode. Figure 4 depicts the internal architecture of Ni-MH batteries.

During discharging, the metal hydride reacts with nickel hydroxide to produce metal (M) and water (H_2O), releasing electrical energy. The discharging reaction is:



Upon charging, the reaction is reversed, converting the metal and water back into metal hydride and nickel hydroxide:



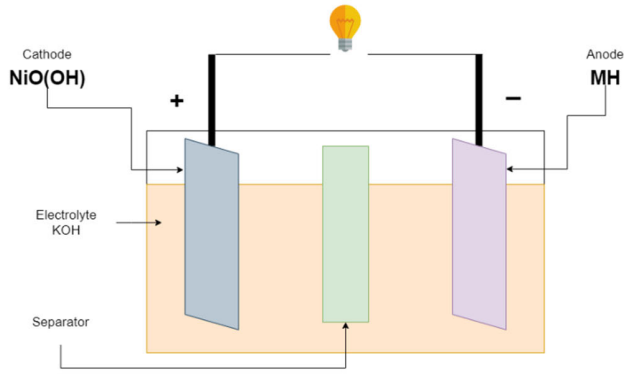


FIGURE 4. The internal architecture of Ni-MH batteries.

4) LI-ION BATTERIES

Lithium-ion batteries are made with a lithium cobalt oxide positive electrode and a graphite negative electrode, with a lithium salt electrolyte. Figure 5 shows the internal architecture of Li-ion batteries.

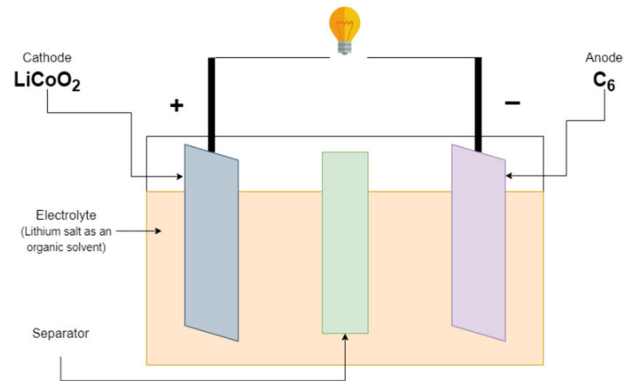
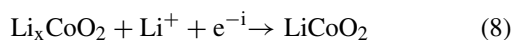


FIGURE 5. The internal architecture of Li-ion batteries.

The nominal voltage of a Li-ion cell is generally between 3.6 and 3.7 volts, and the current capacity varies from a few hundred milliamps to several amps, depending on the application [65]. These batteries are preferred for their high energy density, low self-discharge rate, and long cycle life, making them ideal for portable electronics, electric vehicles, and energy storage systems [66]. Lithium-ion (*Li-ion*) batteries operate through the movement of lithium ions (Li^+) between the electrodes. During discharging, lithium cobalt oxide ($LiCoO_2$) at the positive electrode releases lithium ions and electrons, forming lithium-deficient cobalt oxide (Li_xCoO_2), while the lithium ions move to the negative electrode:



The charging process reverses this reaction, where lithium ions and electrons recombine with lithium-deficient cobalt oxide to reform lithium cobalt oxide:



D. ENERGY SOURCES THEORY

To maximize system performance and sustainability in the field of renewable energy systems, it is essential to comprehend the dynamics of energy generation, storage, and distribution. Mathematical equations play a pivotal role in quantifying these dynamics, providing a thorough framework for analysis and design.

1) WIND TURBINE

Wind turbine power provides a sustainable and renewable energy source by using the kinetic energy of moving air to create electricity. The main parts of these systems are wind turbines (WT), which are made up of blades that harness wind energy to spin a rotor that is coupled to a generator to produce electricity using electromagnetic induction [67].

$$P_{WT} = \frac{1}{2} \rho A V^3 C_p \quad (9)$$

where P_{WT} represents the power extracted from the wind turbine, ρ denotes air density, A signifies the swept area of the turbine blades, V denotes wind speed, and C_p represents the power coefficient reflecting turbine efficiency.

2) PHOTOVOLTAIC SOLAR PANELS

A Photovoltaic (PV) cell is formed by combining p and n type semiconductor material in a thin layer. Therefore, they are similar to diode in structure. As a result, their structures resemble the ones found in diode. The solar radiation is converted into electrical energy by the PV cell. Meanwhile, they fairly generate very low voltage in the high current density [68].

The current that flows when the terminals are shorted together (the short-circuit current, I_{sc}) and the voltage across the terminals when the leads are left open (the open-circuit voltage, V_{OC}) are the two conditions of interest for both the real PV and its equivalent circuit. The ideal current source's magnitude must also equal I_{sc} since the short-circuit current must equal I_{sc} . When the leads of the PV cells are left open, the load V equals $V_{OC} = V_d$ and the load current, I , is null [69]. Figure 6 presents a circuit diagram for single-diode electrical PV module equivalent circuit.

Therefore, a PV cell shows the character of the current source. The relationship between the PV Panel outputs current while considering the losses is expressed by:

$$I = I_{ph} - I_0 \left[\exp \left(\frac{V_{PV} + I_{PV} R_s}{n_s V_t Q_d} \right) - 1 \right] - \frac{V_{PV} + I_{PV} R_s}{R_p} \quad (10)$$

where I represent the current generated by the solar cell, I_{ph} denotes photocurrent generated by the solar irradiance, I_0 signifies diode reverse saturation current, V_{PV} represent voltage across the PV module, I_{PV} signifies the total current flowing through the module, R_s denotes the series resistance, n_s is the number of cells in series, V_t represents thermal voltage, and Q_d signifies the diode ideality factor.

E. ENERGY SOURCES THEORY

The block diagram in Figure 7 below provides a visual representation of the components involved in the hybrid energy system and their interconnections, it helps to illustrate the key elements that researchers have investigated or discussed in their studies related to use of microcontroller for battery discharging and charging of hybrid energy system.

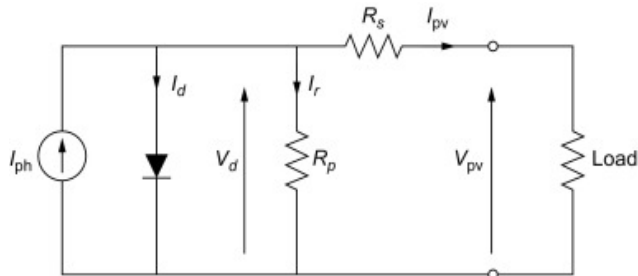


FIGURE 6. Single-diode electrical PV module equivalent circuit.

The interconnected hybrid energy system of renewable energy sources and components facilitates efficient energy generation, storage, and distribution [70]. The main energy generators that use solar and wind power, respectively, are the PV solar panels and wind turbines, which are positioned at the front [71]. These energy sources are fed into a Charge Controller, which controls the charging process while ensuring the Battery Pack's lifespan and optimum performance [72]. The Battery Pack serves as a crucial energy storage unit, storing excess energy generated during peak production periods for later use when energy generation is low [73].

An Inverter with an internal wireless transceiver then converts the stored DC energy from the Battery Pack into AC power, making it compatible with household appliances and other loads [74]. Additionally, the internal wireless transceiver enables seamless communication and monitoring of the system's performance [75]. Finally, the Load, representing various electrical devices and appliances, consumes the converted energy to fulfill power requirements [76]. The integrated system maximizes the use of renewable resources while effectively satisfying the load's demands, ensuring a sustainable and dependable energy supply [77].

F. BATTERY MANAGEMENT SYSTEMS AND MICROCONTROLLER APPLICATIONS

BMS monitors and controls battery parameters such as voltage, current, and temperature to optimize performance and ensure safety [78], [79]. Its functionalities include state-of-charge estimation, cell balancing, and thermal management [80], [81]. Microcontrollers play a vital role in BMS, enabling communication and control within the system [82]. Challenges in BMS development include scalability, system complexity, and accurate modelling for grid-scale applications [83]. Advanced BMS designs incorporate decentralized architectures and self-organizing nodes to improve reliability

and flexibility [82]. Ongoing research focuses on enhancing BMS capabilities to address the growing demands of electric vehicles and renewable energy integration [23].

1) ROLE OF MICROCONTROLLERS IN BATTERY CHARGING AND DISCHARGING

Microcontrollers play a crucial role in battery charging and discharging management, enhancing efficiency, longevity, and safety. Smart charging techniques, including fuzzy logic and model predictive control, optimize charging processes [22], [84]. Microcontroller-based systems monitor key parameters like state-of-charge (SOC), voltage, and temperature to ensure safe operation [23].

Fuzzy-based controllers effectively manage charging and discharging within safe SOC ranges (20-80%), preventing overcharging and over-discharging [85], [86]. These systems can intelligently switch between energy sources in hybrid setups, improving overall efficiency [15]. Centralized controllers assist in managing multiple battery storage units connected to renewable energy sources [21]. The implementation of microcontroller-based battery management systems in electric vehicles and microgrids demonstrates their versatility and importance in various applications [85].

IV. METHODOLOGY

A. RESEARCH TOPIC IDENTIFICATION

The use of microcontroller-driven battery management in hybrid energy systems was identified as a promising research topic. Keywords relevant to this topic were designed and used to search for pertinent documents in the Scopus database. Nevertheless, the authors do acknowledge that using a single database may not capture all relevant literature, as studies have shown limited overlap between major databases [87]. However, this paper integrated Scopus data with VOSviewer software for bibliometric analysis, enabling advanced insights into citation networks, keyword co-occurrence, and emerging research trends, which are particularly useful for systematic reviews.

Hence, the use of multiple databases was not supported in this study, as some database data are not compatible with the used software. Also, Scopus was chosen due to its highly comprehensive database that provides significant benefits for conducting systematic reviews. It is a source of data that covers a wide range of disciplines and geographic regions, making it a valuable resource for diverse research areas [88].

While the above-mentioned features made Scopus an ideal stand-alone database for achieving some of the objectives of this study, it also assisted in managing and narrowing the investigation to a specific focus area. However, future studies can expand to additional databases to further enhance the breadth and depth of analysis. In this work, to ensure comprehensive coverage while using the Scopus database, the selected keywords were "battery discharge" OR "battery charging" AND "hybrid system" OR "Microcontroller".

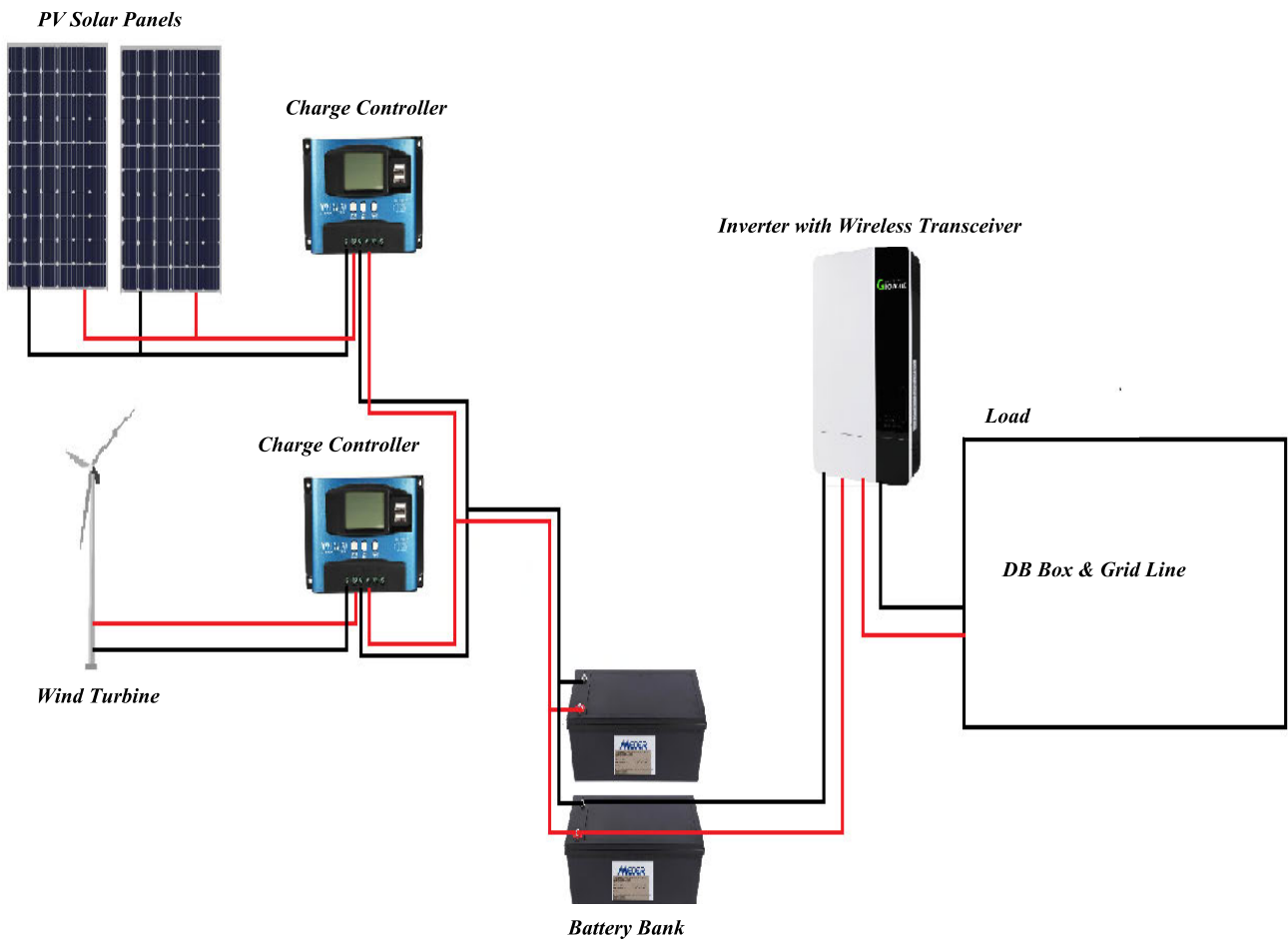


FIGURE 7. A hybrid energy system block diagram: a visual representation of the components involved in the hybrid energy system and their interconnections.

B. DATABASE SEARCH

To ensure relevant results are obtained from Scopus database, the search was set to look within the “Article title, Abstract, Keywords” fields. Keywords were entered into the search bar, and the results were sorted by relevance. To focus on recent developments, the search was restricted to publications from 2018 to 2023. The subject area was filtered to “Engineering,” with document types narrowed to conference and journal papers. Only papers in English were considered. After applying these criteria, 94 articles were identified for further analysis.

C. DATA EXPORT AND INITIAL PROCESSING

The selected papers were exported from Scopus in CSV format, including all relevant fields. The file was downloaded and opened in Microsoft Excel, where the data was transferred into a new workbook titled “Book 1” for further analysis.

D. INCLUSION AND EXCLUSION

Table 1, the inclusion criteria specify the use of Scopus as the database, focusing on article titles, abstracts, and keywords,

with results sorted by relevance. Only articles and conference papers in the field of engineering, published in English between 2018 and 2023, were included, with data exported in CSV format. The exclusion criteria ruled out other databases, non-engineering subjects, non-English publications, older works published before 2018, and irrelevant articles.

V. LITERATURE REVIEW SYNTHESIS

A. MICROCONTROLLER-BASED INNOVATIONS IN BATTERY CHARGING SYSTEMS

Miller et al. in [89] developed a battery-less wireless sensor node powered by Radio Frequency (RF) energy harvesting for off-grid IoT applications utilizing a PIC32MX220F32 microcontroller, they implemented a Constant Current-Constant Voltage (CC-CV) battery charging algorithm. In [90], Djuanda et al. developed an automatic battery charging system based on the Android operating system to prevent overcharging. The system utilizes a microcontroller, Bluetooth module, relay module, and an Android app to control the charging process. Suryadi et al.

Reference [91] designed a controller for a hybrid power plant combining wind and solar energy with a battery as

TABLE 1. Inclusion and exclusion criteria for systematic literature review.

Item No.	Category	Inclusion	Exclusion
1	Database	Scopus	PubMed, Web of Science, WorldCat, etc.
2	Search within	Article title, Abstract, Keyword	All fields, Authors, ISSN, Language, etc.
3	Sort by	Relevance	Date (newest), Title, Type, etc.
4	Document type	Articles, Conference papers	Book, Chapter, Reviews etc.
5	Subject area	Engineering	Computer science, Mathematics, etc.
6	Language	English	French, Chinese, Spanish, etc.
7	File types	CSV	Plain text, RIS, BibTeX, etc.
8	Range	2018-2023	All papers published before 2018
9	Article	Relevant Articles	Irrelevant Articles

storage using the Pulse Width Modulation (PWM) method and ATmega 32 microcontroller. Wu et al. [92] proposed an active cell balancing circuit and charging strategy for lithium battery packs.

Moreover, Ganesh et al. [93] designed and constructed an optimization charging system for Lithium Polymer (Li-Po) batteries using tracked solar panels, implemented in an energy management system for a robotic exploration vehicle featuring a smart host microcontroller for enhanced design flexibility. Then Purwanto et al. [94] developed an auto-cut-off charging system for batteries to maintain battery life by utilizing a Single-Ended Primary-Inductor Converter (SEPIC) converter with a PWM fuzzy logic controller and a relay for system breakers, controlled by an ARMSTM32F407VG microcontroller. As well Santhosh et al. [95] designed and implemented a standalone Photovoltaic (PV) system with a focus on battery charging, using a Programmable Interface Controllers (PIC) microcontroller-based charge controller. Furthermore, Behera et al. [96] Investigated the application of a Dual Active Bridge Converter (DABC) for Electric Vehicle (EV) battery charging, using a C2000 microcontroller.

In 2018 Kevin Siagian et al. [97] developed a battery BMS prototype to supervise battery charging using solar panels, implemented with Arduino Mega. Whereas Kumar et al. [98] designed a standalone hybrid system integrating photovoltaic (PV) and wind energy sources, using an Arduino microcontroller. Suryoatmojo et al. [99], Designed, simulated, and implemented a Synchronous Non-Inverting Buck-Boost (SNIBB) circuit using a four-switch H-bridge and an ATmega 16 microcontroller with a Proportional Integral (PI) controller. To charge the batteries of Electric Vehicles (EVs), Pandey et al. [100] developed a Power Factor Corrected (PFC) boost converter combined with a half bridge Inductor-Inductor-Capacitor (LLC) resonant converter for EV battery charging, based on 2kW prototype using a TMS320F28035 microcontroller. Table 2 presents the summary of microcontroller-based charging systems-related publications.

Using a microprocessor, Jamadar et al. [101] developed a solar-battery hybrid energy source for electric cars. Moreover, Patri et al. [102] designed a battery charging system utilizing solar and utility grid power, implemented with a microcontroller. Similarly, Buntappatan et al. [103]

integrated Wemos D1 Arduino microcontroller and devised a wireless transmission system for real-time monitoring of emergency device batteries. Shiao et al. [104] designed a microcontroller-controlled buck-boost Direct Current to Direct Current (DC-DC) power converter system for Lithium-ion (Li-ion) battery charging. Dhakar et al. [105] also designed a bi-directional CLLC converter and implemented a simple PI-based voltage/current control loop on a 32-bit microcontroller (TMS320F28379D), along with Synchronous Rectification (SR) to control turn-on and turn-off delays on rectifier sides. However, Awal et al. [106] used a microcontroller-based control unit for both the inverter and battery charge controller to create a transformerless 1 kW off-grid inverter with integrated battery charging capability. Also, Gomez et al. [107] created a waste management system that integrated solar and wind-powered microcontroller-controlled charging stations.

B. HYBRID SYSTEMS COMBINING MULTIPLE POWER SOURCES

To maximize the use of the PV array and Doubly Fed Induction Generator (DFIG) wind turbine even under variable load conditions, Boutabba et al. [108] developed an intelligent maximum power point tracking system using fuzzy logic for a standalone hybrid power system. Moreover, to solve the problem of fluctuating currents and voltages from solar panels that can result in overcharging, Taufik Imam Hidayat et al. [109] proposed a hybrid system combining solar panels and batteries, using a multi-input buck converter with Adaptive Neuro-Fuzzy Inference Systems (ANFIS) control.

Pangaribowo et al. [15] developed a control system for hybrid energy systems to manage battery charging and discharging. Vinothkumar et al. [110] also developed an uninterrupted power supply system for hybrid electrical systems using a Luo converter. Additionally, Abadi et al. [111] developed a hybrid charging system integrating a solar tracker and electric grid, employing a flower pollination algorithm optimization-fuzzy logic controller. Similarly, Singh Rajpoot et al. [112] developed a hybrid DC converter that combines two DC sources and an energy storage system (ESS) to power a DC load. An experimental study on a standalone Photovoltaic/Wind/Battery Hybrid system for residential electricity was conducted by Ghanima et al. [113]. Furthermore, Deng et al. [114] proposed a hybrid charging system for EVs

TABLE 2. Summary of microcontroller-based battery charging system innovations.

Ref No.	Authors & Year	Description
[89]	Miller et al. (2020)	Developed a battery-less wireless sensor node powered by RF energy harvesting for off-grid IoT applications using a PIC32MX220F32 microcontroller with a CC-CV battery charging algorithm.
[90]	Djuanda et al (2020)	Developed an automatic battery charging system based on the Android operating system to prevent overcharging, using a microcontroller, Bluetooth module, relay module, and Android app to control the charging process.
[91]	Suryadi et al (2021)	Designed a controller for a hybrid power plant combining wind and solar energy with a battery, using a PWM method and ATmega 32 microcontroller.
[92]	Wu et al (2019)	Proposed an active cell balancing circuit and charging strategy for lithium battery packs, using an MSP430 microcontroller.
[93]	Ganesh et al (2019)	Designed an optimization charging system for Li-Po batteries using tracked solar panels, implemented in an energy management system for a robotic exploration vehicle, and this system was tested on the VANTER robotic platform, featuring a smart host microcontroller.
[94]	Purwanto et al (2022)	Developed an auto-cut-off charging system for batteries to maintain battery life, using a SEPIC converter with a PWM fuzzy logic controller and an ARMSTM32F407VG microcontroller.
[95]	Santhosh et al. (2021)	Designed and implemented a standalone PV system with a focus on battery charging, using a PIC microcontroller-based charge controller.
[96]	Behera et al. 2023	Investigated the application of a dual active bridge converter (DABC) for electric vehicle (EV) battery charging, using a C2000 microcontroller
[97]	Kevin Siagian et al (2018)	Developed a battery management system (BMS) prototype to supervise battery charging using solar panels, implemented with Arduino Mega.
[98]	Kumar Nayak et al (2018)	Designed a standalone hybrid system integrating photovoltaic (PV) and wind energy sources, using an Arduino microcontroller.
[99]	Suryoatmojo et al. (2018)	Designed, simulated, and implemented a SNIBB circuit using a four-switch H-bridge, utilized a microcontroller ATmega 16 with a PI controller
[100]	Pandey et al. (2018)	Developed a PFC boost converter combined with a half bridge LLC resonant converter for EV battery charging, based on 2kW prototype using a TMS320F28035 microcontroller.
[101]	Jamadar et al. (2023)	Developed a solar-battery hybrid energy source for electric vehicles, optimizing solar cell utilization.
[102]	Patri et al. (2018)	Designed a battery charging system utilizing solar and utility grid power, implemented with a microcontroller.
[103]	Buntappatan et al. (2021)	Developed a real-time monitoring system for emergency device batteries using wireless transmission, by means of integrated Wemos D1 Arduino microcontroller with emergency devices to monitor AC input parameters, battery charging status, and battery health
[104]	Shiau et al. (2018)	Designed a microcontroller-controlled buck-boost DC-to-DC power converter system for Li-ion battery charging
[105]	Dhakar et al. (2020)	Designed a bi-directional CLLC converter with an integrated transformer for ESS applications using 48 V batteries.
[106]	Awal et al. (2022)	A simple PI-based voltage/current control loop was implemented on a 32-bit microcontroller (TMS320F28379D), along with synchronous rectification (SR) to control turn-on and turn-off delays on rectifier sides.
[106]	Awal et al. (2022)	Developed a transformerless 1 kW off-grid inverter with integrated battery charging capability, using microcontroller-based control unit for both the inverter and battery charge controller.
[107]	Gomez et al. (2023)	Created a waste management system that integrated solar and wind-powered microcontroller-controlled charging stations

combining both wired and wireless charging modes, whereas Jain et al. [115] devised a simple voltage control strategy to manage power in stand-alone PV battery-based hybrid systems seamlessly. Table 3 presents a summary of hybrid systems combining multiple power sources.

C. CONTROL STRATEGIES AND ALGORITHMS

Abidin et al. [116] designed and validated the use of a fuzzy logic controller as a battery management system for simultaneous charging and discharging processes in a Battery Energy Storage System (BESS). Although, Ram et al. [117] designed and implemented an interleaved DC-DC buck converter for Li-ion battery charging applications, integrating closed-loop control algorithms into a microcontroller. Additionally, Sreenivas et al. [118] developed an algorithm-guided autonomous docking robot with recharging capabilities that can monitor battery voltage and follow a predetermined course.

An Artificial Neural Network (ANN) method for Maximum Power Point Tracking (MPPT) in photovoltaic systems

to maximize output power was proposed by Hidayat et al. [119]. Santhosh et al. [120] also explored methods to maximize solar panel power extraction, focusing on solar panel rotation and MPPT techniques. In 2021, Hadj Dida et al. [121] designed and simulated a photovoltaic DC-DC Boost converter with a Proportional Integral Derivative (PID) controller for satellite energy subsystems. Moreover, Ram et al. [122] designed and implemented a solar PV-based charging system for two and three-wheeler EVs, utilizing interleaved converters and digital control algorithms. To overcome the longer charging period of a piezoelectric battery, Agustinsih et al. [123] implemented a scheduling fuzzy logic method on an Arduino microcontroller using an LTC3588 circuit for energy harvesting. Athavale et al. [124] proposed a compact and lightweight inverter design with high frequency two-stage operation and sine wave output, compatible with various battery chemistries.

But Mubenga et al. [125] addressed capacity variations in used EV batteries by proposing a new class of cell voltage equalizers (EQUs). Moreover, Nistor et al. [126] developed an adaptive method to maximize energy transfer

TABLE 3. Summary of hybrid energy systems integrating multiple power sources.

Ref No.	Authors & Year	Description
[108]	Boutabba et al (2019)	Developed an intelligent maximum power point tracking system using fuzzy logic for a standalone hybrid power system.
[109]	Taufik Imam Hidayat et al (2022)	Proposed a hybrid system combining solar panels and batteries, using a multi-input buck converter with ANFIS control.
[15]	Pangaribowo et al (2021)	Developed a control system for hybrid energy systems to manage battery charging and discharging.
[110]	Vinothkumar et al (2018)	Developed an uninterrupted power supply system for hybrid electrical systems using a Luo converter.
[111]	Abadi et al (2022)	Developed a hybrid charging system integrating a solar tracker and electric grid, employing a flower pollination algorithm optimization-fuzzy logic controller (FPAO-FLC).
[112]	Singh Rajpoot et al (2022)	Developed a hybrid DC converter that combines two DC sources and an energy storage system (ESS) to power a DC load.
[113]	Ghanima et al (2020)	Conducted an experimental study on a standalone Photovoltaic/Wind/Battery Hybrid system for residential electricity.
[114]	Deng et al (2020)	Proposed a hybrid charging system for EVs combining both wired and wireless charging modes.
[115]	Jain et al. (2021)	devised a simple voltage control strategy to manage power in stand-alone photovoltaic (PV) battery-based hybrid systems seamlessly

to a car battery during regenerative braking by utilizing a battery management algorithm implemented on an ATMEGA 328 microcontroller and an MC34063 DC-To-DC converter control circuit. Corti et al. [127] developed a model-based approach for effective power supply control in a wireless sensor network by using Pulse-Skipping Modulation to control current when a solar source charges a battery. However, Sahoo et al. [128] introduced a novel method for battery charging and motor control in EVs, using two highly efficient DC-DC converters called Buck-Boost converter. Table 4 presents a summary of control strategies and algorithms related publications.

As well as Kang et al. [129] proposed a control method for simultaneous battery charging and load-following in combustion engine-based gasoline electric power systems for multi-copter drones. Drumea et al. [130] concentrated on modelling and simulating an energy consumption of an autonomous embedded system powered by a solar cell charging, Nickel–Metal Hydride (NiMH) battery charging and discharging, and microcontroller operation. Similarly, Utomo et al. [131] used the Incremental Conductance (IC) algorithm for MPPT to address the lack of wind speed capacity and photon emission in a hybrid PV-wind system. On the other hand, to get beyond the drawbacks of traditional chargers, Banda et al. [132] created an ANN controlled Synchronous Buck (SB) converter for lead-acid battery charging. Additionally, Shanmugam et al. [133] proposed an Integrated Three Port Converter (ITPC) and B4-Inverter Brushless Direct Current Motor (BLDC) drive targeting low or medium applications. Yukhalang et al. [134] designed a battery charging control system for a stand-alone wind-driven Self-Excited Induction Generator (SEIG).

D. RENEWABLE ENERGY-BASED CHARGING SYSTEMS

Choifin et al. [135] Designed and implemented a charger controller for a 6-volt, 4.5 Ah battery load using a mini wind turbine plant, along with a pulse signal generator and an

ARM Cortex M0-NUC 120 Microcontroller. Using renewable energy sources (Wind-PV-FC-Battery), El-Naggar et al. [136] designed an ideal power control and energy management approach for Plug-In Electric Vehicle (PEV) charging stations.

Furthermore, using a microprocessor and sensors, Asrori et al. [137] investigated the charging efficiency of a 100 Wp solar panel integrated into an electric bicycle and presented the results on a Liquid Crystal Display (LCD) screen. However, Praveen et al. [138] designed and developed a solar-based converter to regulate the solar panel voltage to match the battery voltage, incorporating MPPT for optimal solar panel output. Table 5 presents a summary of renewable energy-based charging systems-related publications.

VI. RESULTS AND DISCUSSIONS

A. PATTERNS IDENTIFICATION ANALYSIS

The research on microcontroller-based battery management systems showcases exciting innovations but also highlights a few areas where improvements are needed. A common theme is the flexibility of microcontrollers like Arduino Mega, PIC32MX220F32, and ARMSTM32F407VG. These devices have been used in everything from small project setups to advanced industrial systems. For instance, Arduino is a favorite for its affordability and simplicity, but it can be limiting for more complex tasks. On the other hand, high-performance controllers like TMS320F28379D are great for implementing advanced algorithms but might feel like overkill for smaller projects.

One of the most interesting trends is how these systems are tied to renewable energy. Researchers are combining solar, wind, and batteries to create hybrid setups that maximize energy use. Tools like MPPT and PWM ensure these systems squeeze every drop of energy from renewable sources. This is a great step forward, but challenges remain available e.g., inconsistent power output from solar or wind, which can make charging batteries tricky. Some studies have tackled

TABLE 4. Summary of control strategies and algorithms for battery management systems.

Ref No.	Authors & Year	Description
[116]	Abidin et al. (2022)	Designed and validated the use of a fuzzy logic controller as a battery management system for simultaneous charging and discharging processes in a battery energy storage system (BESS).
[117]	Ram et al. (2023)	Designed and implemented an interleaved DC-DC buck converter for Lithium-ion battery charging applications, integrating closed-loop control algorithms into a microcontroller.
[118]	Sreenivas Rao et al. (2021)	Implemented an automatic docking robot with recharging capability, utilizing IR sensors and Bluetooth for data transmission.
[119]	Hidayat et al. (2019)	Proposed an ANN method for MPPT in photovoltaic systems to maximize output power
[120]	Santhosh et al. (2022)	Explored methods to maximize solar panel power extraction, focusing on solar panel rotation and MPPT techniques.
[121]	Hadj Dida et al. (2021)	Designed and simulated a photovoltaic DC-DC Boost converter with a PID controller for satellite energy subsystems. (
[122]	Ram et al. (2023)	Designed and implemented a solar PV-based charging system for two and three-wheeler EVs, utilizing interleaved converters and digital control algorithms.
[123]	Agustiningih et al. (2018)	Utilized an LTC3588 circuit for energy harvesting to address prolonged charging time of a piezoelectric battery accumulator.
[124]	Athavale et al. (2020)	Proposed a compact and lightweight inverter design with high frequency two-stage operation and sine wave output, compatible with various battery chemistries.
[125]	Mubenga et al (2019)	Addressed capacity variations in used EV batteries by proposing a new class of cell voltage equalizers (EQUs)
[126]	Nistor et al (2022)	Developed an adaptive method to maximize energy transfer to a car's battery during regenerative braking by utilizing a battery management algorithm implemented on an ATMEGA 328 microcontroller and a MC34063 DC-To-DC converter control circuit.
[127]	Corti et al. (2022)	Developed a model-based strategy for efficient power supply control in a wireless sensor network, using Pulse-Skipping Modulation.
[128]	Sahoo et al (2021)	Introduced a novel method for battery charging and motor control in EVs, using two highly efficient DC-DC converters called Buck-Boost converter.
[129]	Kang et al (2023)	Proposed a control method for simultaneous battery charging and load-following in combustion engine-based gasoline-electric power systems for multi-copter drones.
[130]	Drumea et al (2020)	Focused on modelling and simulating energy consumption of an autonomous embedded system powered by a solar cell charging, NiMH battery charging and discharging, and microcontroller operation.
[131]	Utomo et al (2022)	Addressed lack of wind speed capacity and photon emission in a hybrid PV-wind system by applying the incremental conductance (IC) algorithm for MPPT.
[132]	Banda et al (2022)	Designed an ANN-controlled synchronous buck converter for charging lead-acid batteries.
[133]	Shanmugam et al (2020)	Designed and implemented an improved three-interface converter and B4-Inverter fed brushless direct current motor drive for industrial applications.
[134]	Yukhalang et al (2019)	Designed a battery charging control system for a stand-alone wind-driven self-excited induction generator (SEIG).

TABLE 5. Summary of renewable energy-based charging systems.

Ref No.	Authors & Year	Description
[135]	Choifin et al (2019)	Designed and implemented a charger controller for a 6-volt, 4.5 Ah battery load using a mini wind turbine plant.
[136]	El-Naggar et al (2018)	Designed an ideal power control and energy management approach for Plug-In Electric Vehicle (PEV) charging stations
[137]	Asrori et al (2023)	Investigated the charging efficiency of a 100 Wp solar panel integrated into an electric bicycle and presented the results on a Liquid Crystal Display (LCD) screen.
[138]	Praveen et al. (2022)	Designed and developed a solar-based converter to regulate the solar panel voltage to match the battery voltage, incorporating MPPT for optimal solar panel output.

this with smart algorithms like Incremental Conductance or even fuzzy logic, which work well but are not always easy or cheap to implement. Interestingly, not all the research agrees on the best way forward. ANN-based methods are powerful and precise but can be resource-hungry, making them harder to justify in simpler setups. Meanwhile, fuzzy logic is easier to use but might not handle complex conditions as effectively. Even the choice of batteries stirs debate. Li-ion batteries are efficient and long-lasting, but lead-acid options still get attention for their low cost, even if they do not last as long.

Despite the progress, there is plenty of room to grow. Most of these projects focus on small-scale systems, so there is little insight into how they work in larger industrial setups. And while the technologies look promising in labs, it is not yet known how these systems hold up over years of real-world use. There is also a noticeable lack of standard rules for combining all these technologies, which could make scaling them up even harder. Some researchers are starting to explore the potential of IoT for predictive maintenance, but those ideas are still in their infancy. Therefore, microcontroller-based systems for battery management have come a long way,

offering efficient, renewable-focused solutions. But to truly make an impact, there is a need to figure out how to scale them up, standardize their use, and test them in real-world conditions. Future research could unlock their full potential and pave the way for smarter, greener energy systems.

B. THEMES AND EMERGING TRENDS ANALYSIS

1) KEY CONCEPTS ANALYSIS

Figure 8 shows the relationships between key ideas in microcontroller-driven battery management systems for hybrid energy solutions. The diagram uses circles, connecting lines, and colors to help visualize these connections and show which concepts are most important. The circles (nodes) represent different concepts. Bigger circles, like “secondary batteries” and “charging (batteries),” highlight the main topics, while smaller circles, such as “buck converter” and “solar concentrators,” represent more specific or emerging areas of focus.

The lines connecting the circles show how concepts are related. Thicker and denser lines indicate strong relationships, like those seen around “secondary batteries.” Thinner or fewer lines suggest weaker connections, which could mean these areas are less studied or newer in the research space.

The colors divide the circles into clusters based on themes. The green cluster focuses on battery-related technologies, including charging systems and converters. The red cluster emphasizes microcontrollers and system design. The blue cluster groups renewable energy topics, like solar and wind power, along with advanced methods like fuzzy logic. Finally, the yellow cluster highlights solar energy technologies, such as photovoltaics and charge controllers. The overall network shows which topics are central and which are on the edges. Central nodes, like “secondary batteries,” are heavily linked, showing they play a key role in connecting other ideas, such as renewable energy and control systems. Less connected nodes, like “efficiency,” point to specialized or less-explored areas. In summary, Figure 8 helps map out the key areas and connections in microcontroller-driven battery management. It shows the main focus areas, highlights important links, and reveals opportunities for further exploration in this field.

2) TREND ANALYSIS

Figure 9 shows how research into microcontroller-driven battery management systems has shifted over time. In 2018 and 2019, much of the focus was on energy storage and designing systems to support microcontrollers. This early work was about getting the basics right, and figuring out how these systems could function reliably. By 2020 and 2021, attention started moving toward renewable energy sources. Solar and wind power became popular topics, and researchers explored how batteries could be better integrated with these energy systems. There was also growing interest in methods like fuzzy logic to make these systems more efficient and easier to control.

In the final two years, 2022 and 2023, the focus turned to fine-tuning technologies. Topics like MPPT gained prominence as researchers worked on ways to extract the most energy from solar panels. There was also a lot of effort put into improving photovoltaic systems and minimizing energy losses. This timeline paints a clear picture of progress. Researchers moved from setting up the basics to tackling renewable energy integration and, finally, to creating smarter, more efficient systems.

3) THEMES AND DENSITY ANALYSIS

Table 6 organizes research topics into five clusters, showing how these areas are connected and how much attention they have received in studies. The table highlights metrics like the number of links, the strength of those connections, how often topics appear, when they were published, and how frequently they have been cited. The red cluster focuses on topics like battery management systems and renewable energy. Battery management systems have 25 links, a total link strength of 47, and 8 occurrences, making them a core part of this research. MPPT, with 27 links and a similar link strength, reflects its importance in improving energy use in renewable systems. Though renewable energy resources appear less frequently (3 occurrences), their high average citation count (9.33) suggests that early research in this area has been influential.

The green cluster centers on charging systems and secondary batteries. Charging (batteries) stands out with 40 links, a total link strength of 140, and 22 occurrences, making it the most studied topic in this cluster. Secondary batteries follow closely, with 37 links and a link strength of 129, emphasizing their importance in energy storage. Most research in this cluster was published around 2020, showing that these are active areas of study. The blue cluster is about control systems and optimization. Controllers have 36 links, a total link strength of 104, and 14 occurrences, showing their importance in managing hybrid systems. Techniques like MPPT (30 links, total link strength of 77) and fuzzy logic (7.75 average citations) highlight efforts to make systems more efficient and effective.

The yellow cluster focuses on solar energy and energy conversion technologies. DC-DC converters, with 33 links and a total link strength of 71, and solar energy, with 24 links and a link strength of 43, are key topics. With an average publication year of 2021, this cluster reflects newer work aimed at integrating solar power into hybrid systems. The purple cluster explores hybrid systems and efficiency. Hybrid systems have 28 links, a total link strength of 45, and 9 occurrences, showing their role in combining energy sources. Efficiency appears only three times and has no citations, suggesting this is a less developed area with room for further research. Therefore, the table highlights the focus on battery management, charging systems, and control strategies while showing increasing interest in solar energy. Topics like efficiency remain underexplored but offer opportunities for future work. These metrics help paint a clear picture of the current state of research and where it might be headed next.

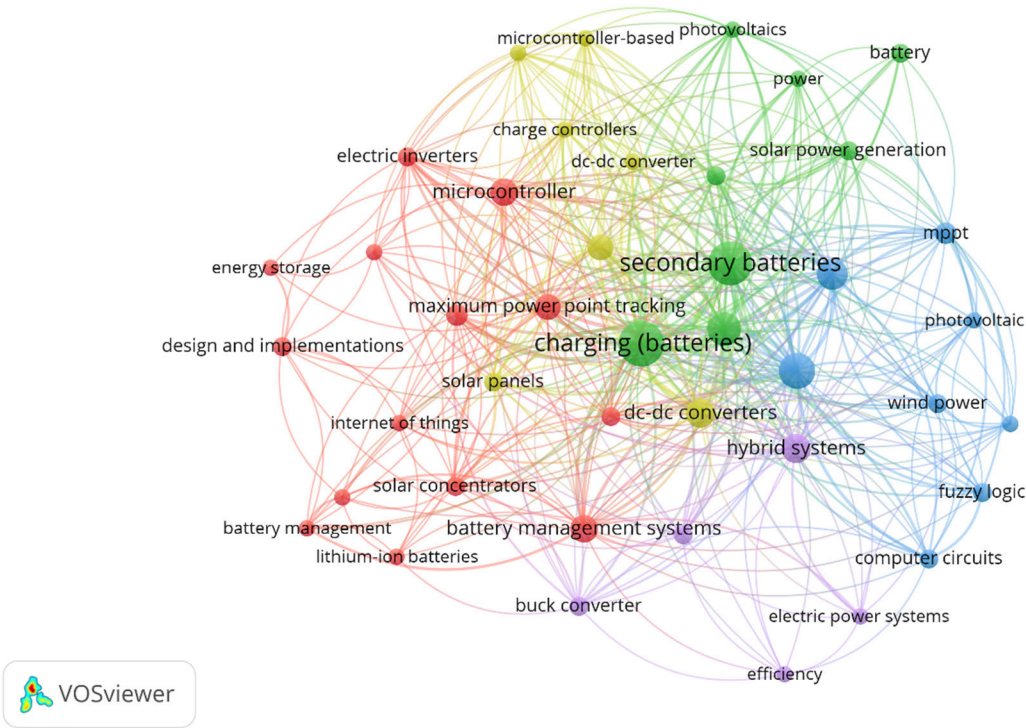


FIGURE 8. Network visualization of key concepts in microcontroller-driven battery management systems.

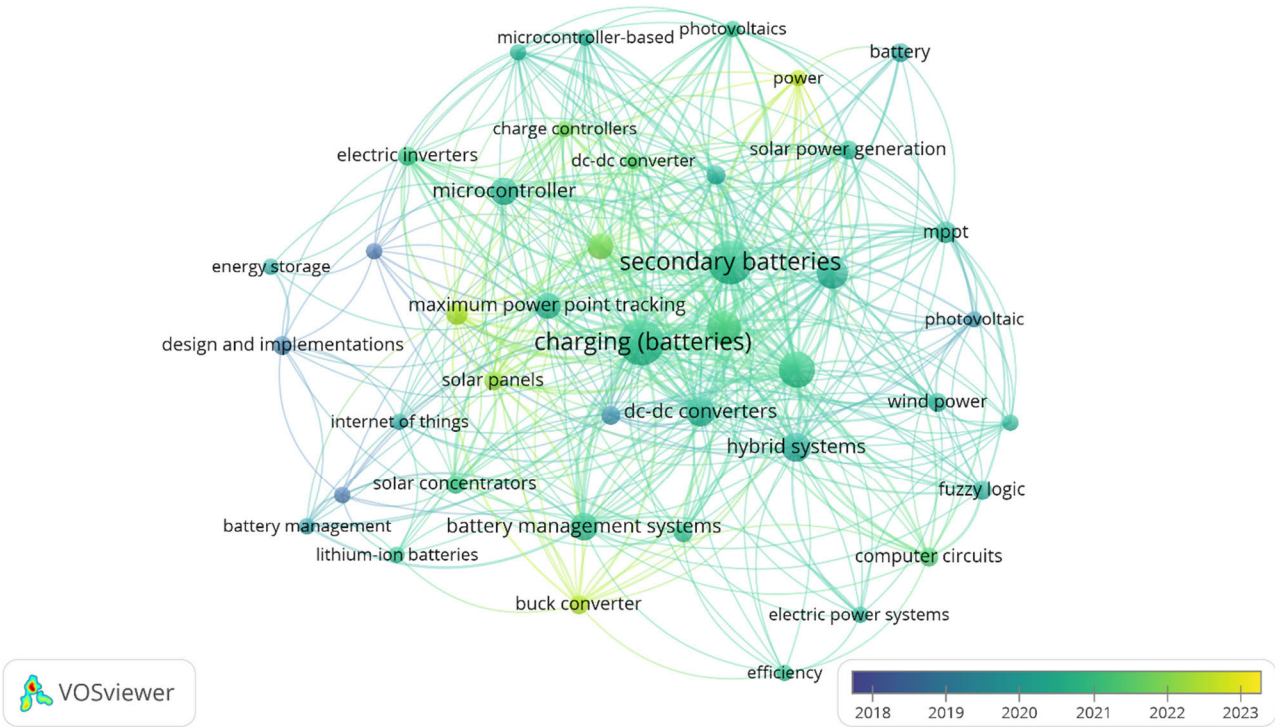


FIGURE 9. Keywords network and emerging trends (2018–2023) in microcontroller-driven battery management systems.

Figure 10 provides a visual map of how research topics are distributed in the field of microcontroller-driven battery management. Brighter areas represent topics that researchers have

focused on the most, while darker areas indicate topics that have received less attention. This helps to see which areas are well-studied and which might need further exploration. The

TABLE 6. Keyword clusters and research metrics in microcontroller-driven battery management systems.

Cluster 1 (Red)					
Keywords	Links	Total link strength	Occurrences	Avg. pub. year	Avg. citations
battery charging	21	37	5	2022.2	1.6
battery management	11	15	3	2019.6667	4
battery management systems	25	47	8	2020.625	2.125
battery pack	9	12	3	2018.6667	6
design and implementations	11	12	4	2019	3.75
electric inverters	22	32	4	2021	5
energy storage	5	6	3	2020	5.3333
internet of things	12	13	3	2020	2
lithium-ion batteries	9	9	3	2020.6667	5
maximum power point tracking	27	47	7	2020.5714	5.4286
microcontroller	24	43	8	2020.5	5.125
photovoltaic cells	19	23	4	2019.25	2.75
renewable energy resources	13	15	3	2018.6667	9.3333
solar concentrators	20	30	4	2021	2
Cluster 2 (Green)					
Keywords	Links	Total link strength	Occurrences	Avg. pub. year	Avg. citations
battery	9	12	4	2020	2.75
charging (batteries)	40	140	22	2020.7273	3.3636
microcontrollers	33	93	12	2021.0833	2.75
photovoltaics	20	31	3	2020.6667	7
power	16	21	3	2022.3333	1.6667
secondary batteries	37	129	20	2020.7	3.4
solar power generation	23	37	4	2020.5	5.5
timing circuits	17	28	4	2020.25	4.25
Cluster 3 (Blue)					
Keywords	Links	Total link strength	Occurrences	Avg. pub. year	Avg. citations
computer circuits	15	26	4	2021.25	1.75
controllers	36	104	14	2020.8571	2.8571
fuzzy logic	14	22	4	2020.25	7.75
maximum power point trackers	30	77	10	2020.4	4.6
mppt	22	36	5	2020.4	4.8
photovoltaic	16	22	3	2019.3333	3
wind power	20	29	4	2020.25	1.25
wind turbine	14	15	3	2020.3333	1.3333
Cluster 4 (Yellow)					
Keywords	Links	Total link strength	Occurrences	Avg. pub. year	Avg. citations
charge controllers	21	33	3	2021.6667	6
dc-dc converter	19	28	3	2021.3333	5.3333
dc-dc converters	33	71	9	2020.4444	4.5556
microcontroller-based	20	28	3	2020.3333	5.6667
photovoltaic systems	20	28	3	2020.3333	5.6667
solar energy	24	43	7	2021.7143	4.2857
solar panels	20	35	4	2022	2.25
Cluster 5 (Purple)					
Keywords	Links	Total link strength	Occurrences	Avg. pub. year	Avg. citations
buck converter	17	24	4	2022.25	1.25
efficiency	10	10	3	2020.6667	0
electric machine control	19	26	4	2020.75	4.5
electric power systems	11	14	3	2020.3333	4.3333
hybrid systems	28	45	9	2020.2222	5.3333

brightest areas of the figure focus on secondary batteries and charging (batteries). These are clearly the most studied topics, reflecting their importance in energy storage and charging systems within hybrid energy applications. Surrounding these core areas, terms like battery management systems,

MPPT and dc-dc converters are observed. These are critical technologies that support battery performance and energy management. Their position near the brightest areas shows their strong connection to the main focus of research. Further out, areas with medium brightness include terms like solar power generation, photovoltaic, and solar panels.

These topics reflect ongoing work on integrating renewable energy into battery-driven systems. Nearby keywords such as microcontroller and charge controllers emphasize the role of technology in supporting renewable energy systems.

In the darker regions, topics like efficiency and buck converter are observed, indicating less frequent studies, suggesting they offer opportunities for further research. Terms like fuzzy logic and wind power are also present in the outer areas, indicating niche topics that may still hold potential for deeper exploration.

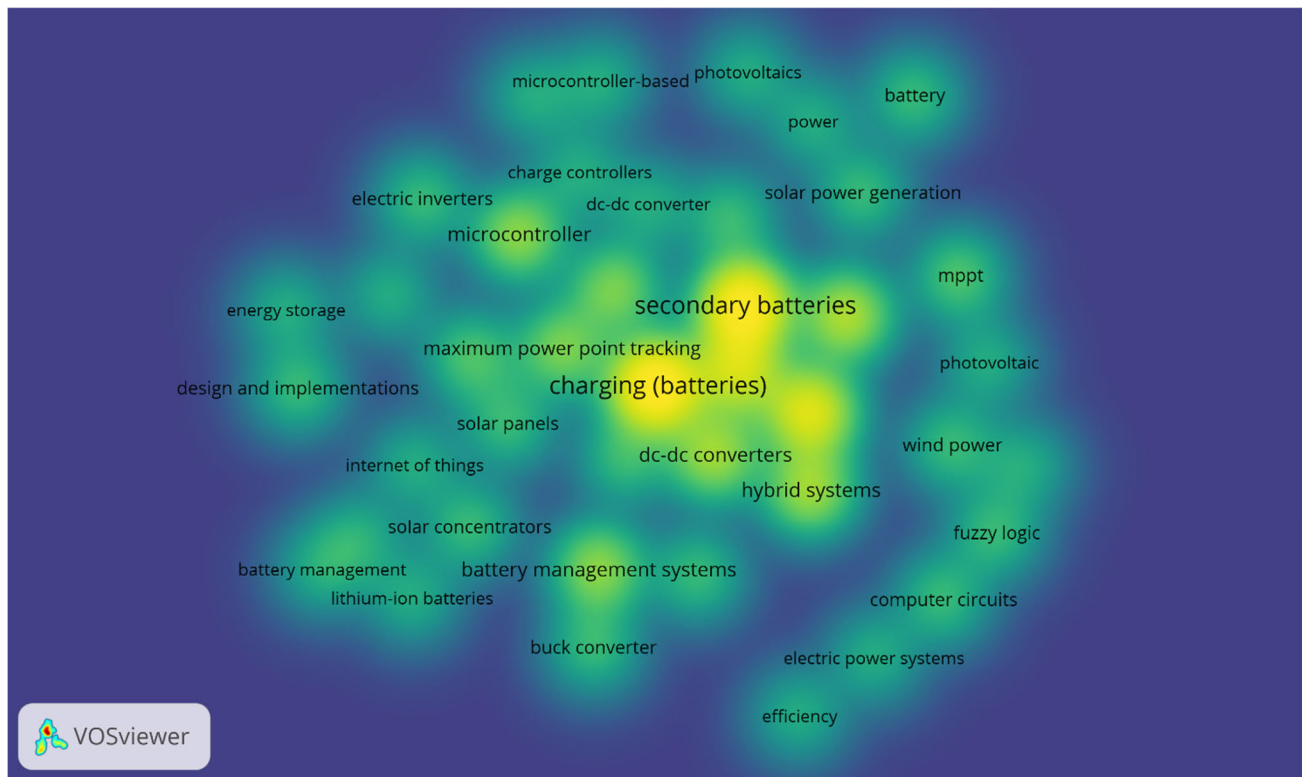


FIGURE 10. Density visualization of research focus in microcontroller-driven battery management system.

Figure 10 shows that research is heavily focused on batteries, charging systems, and technologies supporting energy management. While renewable energy integration is a growing area, less attention has been given to efficiency and system-level optimization. This map gives a clear view of where the research is concentrated and points to possible gaps that could be addressed in the future.

VII. GAP IDENTIFICATION AND FUTURE RESEARCH DIRECTIONS

A. GAP IDENTIFICATION

The research on microcontroller-based battery management systems shows some clear gaps. These come from both the technical challenges and the way the research has been prioritized. One problem is that the microcontrollers being used do not always fit the task. For example, Arduino Mega is cheap and simple but struggles with complex systems. On the other hand, advanced options like the TMS320F28379D are expensive and often more powerful than what is needed for smaller projects. This creates a gap where mid-range options could be very useful. Another issue comes from renewable energy sources like solar and wind. They do not produce steady power, which makes managing batteries harder. While algorithms like fuzzy logic help with this, they are expensive and not easy to apply everywhere.

There is also a lack of agreement in the research. Advanced methods like ANN are precise but require too many resources for smaller systems. Simpler options, like fuzzy logic, are

easier to use but do not always perform well in complex situations. On top of this, the choice of batteries is not clear. Li-ion batteries are long-lasting and efficient but expensive. Lead-acid batteries are cheaper but do not last as long. Without clear guidelines, it is hard to decide which is best in different situations. Most research is also done on small-scale systems, so we do not know how these ideas will work in bigger industrial setups. There are also no clear rules for how to combine the different technologies, which makes scaling up even harder.

Looking at the trends, Figure 8 shows that topics like secondary batteries and charging systems get most of the attention. Less studied areas, like efficiency, seem to be overlooked. Figure 9 shows that research started with basic designs and renewable integration. More recently, the focus has been on improving tools like MPPT and photovoltaic systems. However, there is not much testing in real-world conditions, which makes it hard to know how these systems will work outside the lab. The analysis in Table 6 shows the same imbalance. Research focuses a lot on battery management and renewable energy integration. Topics like hybrid system efficiency or advanced converters, like buck converters, do not get as much attention. This suggests some areas with potential are being ignored.

B. FUTURE RESEARCH DIRECTIONS

Future research needs to focus on making microcontrollers better suited for different tasks. Right now, affordable options

like Arduino work well for basic projects but are not powerful enough for more complex systems. High-end models like the TMS320F28379D are too expensive and unnecessary for smaller setups. A balance between cost and performance is missing, and filling this gap could help many applications. Renewable energy, like solar and wind, brings challenges because its output is inconsistent. This makes it hard to charge batteries effectively. Current solutions, like fuzzy logic, help manage these fluctuations but are often too complicated or costly for wider use. Research should work on simpler, cheaper tools that handle energy changes more efficiently. Another issue is that there are no clear guidelines for putting these systems together. Microcontrollers, batteries, and renewable energy systems need to work seamlessly, but without a standard approach, building reliable systems is difficult. A straightforward framework would make it easier to scale these setups for larger projects.

Most testing is done in labs, which does not show how these systems perform in real-world conditions. Researchers need to focus on testing these setups in unpredictable environments to understand how they work overtime and in larger applications. Some areas have not been explored enough. Improving energy efficiency or using advanced converters, like buck converters, could make systems more effective, but they are not getting enough attention. Hybrid systems, which combine different energy sources, also need more research to figure out the best designs. Battery selection is another question. Li-ion batteries are efficient and last longer, but they are expensive. Lead-acid batteries are cheaper but do not perform as well over time. Clear advice on which batteries work best for specific setups would help simplify system design. Renewable energy tools like MPPT need improvement too. They are helpful but do not handle sudden changes in energy supply well. Systems that can adapt automatically to these fluctuations would make setups more reliable and stable. IoT technology could make a big difference, i.e., sensors that monitor battery health and system performance in real time could identify problems early, helping reduce downtime and improve reliability. Blockchain and machine learning could also add new capabilities. Blockchain might make energy trading easier, while machine learning could predict energy needs and optimize how systems run.

VIII. CONCLUSION

This study performed a systematic review to explore applications, control strategies, and emerging trends on microcontroller-driven battery management in hybrid energy systems. Using a structured approach, relevant documents were gathered through targeted keyword searches in the Scopus database, focusing on recent engineering research. The selected articles were then refined, exported, and thoroughly reviewed for relevance, resulting in a robust dataset for analysis. The key findings from the literature review highlighted that microcontroller-based systems are making big strides in renewable energy and battery management. Tools like MPPT and PWM help improve efficiency, while microcontrollers

like Arduino and TMS320F28379D balance cost and performance depending on the project. Lithium-ion batteries are favoured for their long life, but lead-acid batteries still have their place because they are cheaper. Recent research is focusing on making charging systems faster and more efficient, while older studies built the basics for energy storage and management. Newer ideas like IoT could make these systems smarter, but challenges like scaling up, standardization, and long-term reliability remain. Solving these issues will be key to create practical, and sustainable energy systems. The study successfully achieved its key objectives, offering a clear framework, exploring relevant methodologies, and synthesizing critical findings. It also identified important gaps for future research, providing valuable insights to the field and setting the stage for further advancements.

ACKNOWLEDGMENT

The authors convey their gratitude and gratefulness to the Institution.

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